FAIR 3D HERITAGE
Documentation, Publication and Preservation of the Digital 3D Heritage on the Web

When More Data Is Not Enough: Preserving Robotically Fabricated Architecture

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Abstract

The increasing accumulation of 3D data in the fields of cultural heritage and the built environment has intensified the need for documentation practices that support not only visualisation and dissemination, but also long-term preservation and reuse. While conventional heritage documentation has often addressed the problem of missing or fragmentary information, robotically fabricated architecture represents the opposite condition: a surplus of data (Halpern et al. 2022). This refers not only to quantitative concerns about data generation but also to the qualitative contexts both within and between datasets. If such abundance risks dissolving the built object into a field of data correlations, preservation can reinforce causation by relating design, fabrication, material behaviour, and subsequent change.

This paper addresses this intent within the framework of the Digital Construction Archive project, which investigates preservation strategies for digitally fabricated architecture whose identity cannot be separated from its digital design and fabrication processes, represented in the term "digital materiality" (Gramazio and Kohler 2008). The project brings together complementary focuses on discrete assembly and amorphously joined material systems, allowing preservation questions to be examined across different logics of construction. In this context, preservation cannot be limited to the built artefact alone, but must include the archiving of fabrication and monitoring data, file structures, and workflow knowledge that sustain its long-term accessibility and repairability following the principles of the Open Archival Information System. By addressing complementary phases of preservation through documentation, monitoring, and upkeep, the project establishes a replicable workflow for long-term stewardship and provides new preservation guidelines for digitally fabricated heritage.

Within this broader framework, the multimodal scanning of Tor Alva, a 3D-printed reinforced concrete tower, serves as the monitoring component of the research. Through

terrestrial laser scanning, photogrammetry, and structured data curation, the study establishes a baseline record of the completed artefact and supports repeated observation of condition and material change to predict material deformation before structural failures manifest. In this way, monitoring is not limited to the retrospective documentation of deterioration, but contributes to the anticipation of damage and the development of maintenance and repair logics for a still-emerging construction method. The resulting 3D representations and associated data structures are intended to be made accessible as semantically enriched smart pointclouds via the Digital Construction Archive, a web-based gated repository which uses semantic web technology to preserve critical data for future repair scenarios. The archive employs the PREMIS metadata standard to support FAIR and TRUST principles by using interoperable data structures, while manual filtering and semantic curation, achieved through close consultation with project stakeholders, ensure the quality of the information. This raises questions of authenticity that extend beyond the built artefact itself, encompassing the data and the authority by which future interventions are legitimised. In an environment of surplus data, established guidelines must therefore be reconsidered to support curatorial decision-making for long-term preservation.

References

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